

Palm Beach State College • Spring 2026

Unit 03 Study Notes

MAP2302 Differential Equations

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Examination: Friday, April 3rd, 2026

Topics covered: Power Series Solutions · Variation Of Parameters · Applications of ODEs Across Disciplines ·

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Appendix: Quick Reference

A.1 Integration by Parts Shortcuts

Recall the IBP formula: $\int u dv = uv - \int v du$. For repeated application use the **tabular (tic-tac-toe) method**: alternate signs $+, -, +, -, \dots$, differentiate the left column to zero, integrate the right column, then multiply diagonally.

Integral	Result	Notes
$\int x e^{ax} dx$	$\frac{e^{ax}}{a} \left(x - \frac{1}{a} \right) + C$	$u = x, dv = e^{ax} dx$
$\int x^2 e^{ax} dx$	$\frac{e^{ax}}{a} \left(x^2 - \frac{2x}{a} + \frac{2}{a^2} \right) + C$	tabular method
$\int x e^{ax} \cos(bx) dx$	use tabular twice	cycles after 2 steps
$\int e^{ax} \cos(bx) dx$	$\frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2} + C$	IBP twice, solve for I
$\int e^{ax} \sin(bx) dx$	$\frac{e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2} + C$	IBP twice, solve for I
$\int x \cos(bx) dx$	$\frac{\cos(bx)}{b^2} + \frac{x \sin(bx)}{b} + C$	$u = x, dv = \cos(bx) dx$
$\int x \sin(bx) dx$	$\frac{\sin(bx)}{b^2} - \frac{x \cos(bx)}{b} + C$	$u = x, dv = \sin(bx) dx$
$\int x^n e^{ax} dx$	$\frac{x^n e^{ax}}{a} - \frac{n}{a} \int x^{n-1} e^{ax} dx$	reduction formula

A.2 Trigonometric Identities

Reciprocal and Quotient Identities

$$\csc x = \frac{1}{\sin x}, \quad \sec x = \frac{1}{\cos x}, \quad \cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x}, \quad \tan x = \frac{\sin x}{\cos x}.$$

Pythagorean Identities

$$\sin^2 x + \cos^2 x = 1, \quad 1 + \tan^2 x = \sec^2 x, \quad 1 + \cot^2 x = \csc^2 x.$$

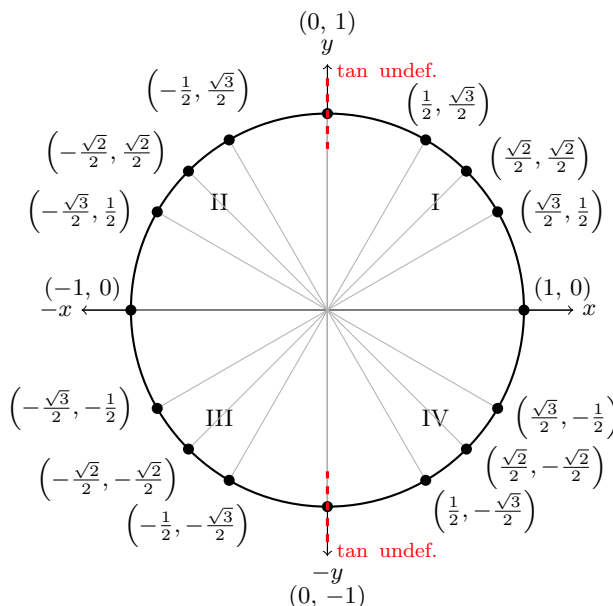
Double-Angle Identities

$$\sin(2x) = 2 \sin x \cos x, \quad \cos(2x) = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x = 2 \cos^2 x - 1.$$

Power-Reduction Identities

$$\sin^2 x = \frac{1 - \cos(2x)}{2}, \quad \cos^2 x = \frac{1 + \cos(2x)}{2}.$$

A.3 Unit Circle and Domain Restrictions



Why $\tan x$ causes problems at $x = \pm\pi/2$. Since $\tan x = \sin x / \cos x$, the function is undefined wherever $\cos x = 0$, namely at $x = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \dots$. On the unit circle these correspond to the points $(0, 1)$ and $(0, -1)$ where the x -coordinate (i.e. $\cos x$) is zero.

For an ODE such as $y'' + (\tan x)y = e^x$, the coefficient $\tan x$ is **continuous on any open interval not containing these points**. The largest such interval centered at $x = 0$ is therefore $(-\frac{\pi}{2}, \frac{\pi}{2})$, which is exactly the interval guaranteed by the existence and uniqueness theorem.

Function	Undefined at	Continuous on
$\tan x = \sin x / \cos x$	$x = \frac{\pi}{2} + n\pi$	each $(-\frac{\pi}{2} + n\pi, \frac{\pi}{2} + n\pi)$
$\cot x = \cos x / \sin x$	$x = n\pi$	each $(n\pi, (n+1)\pi)$
$\sec x = 1 / \cos x$	$x = \frac{\pi}{2} + n\pi$	same as $\tan x$
$\csc x = 1 / \sin x$	$x = n\pi$	same as $\cot x$

$n \in \mathbb{Z}$

1 Variation Of Paramters

1.1 Application to linear nth-order systems

1.1.1 Application to Linear 1st order systems

1.1.2 Application to Linear 2nd order systems

1.2 Introduction of Integral-Defined Functions

1.3 Discussing the Role of The Constant of Integration

2 Cauchy-Euler Equations

2.1

2.1.1 General Form and Terminology

3 Spring - Mass Systems

3.1 Introduction to Higher Order Linear ODEs

3.1.1 General Form and Terminology

4 Power Series Solutions

4.1 Introduction to Higher Order Linear ODEs

4.1.1 General Form and Terminology

Practice Problems

One problem of each exam type from Unit 02. Work through these at the end.

P1. *Existence & Uniqueness — Interval of Solution*

Find the largest interval centered about $x = 0$ on which the IVP has a unique solution.

$$y'' + (\tan x)y = e^x, \quad y(0) = 1, \quad y'(0) = 0.$$

P2. *Wronskian and Fundamental Set Verification*

Verify that $y_1 = x^2$ and $y_2 = x^4$ form a fundamental set of solutions of $x^2y'' - 5xy' + 8y = 0$ on $(0, \infty)$ by computing $W(x^2, x^4)$.

P3. *Reduction of Order*

Given that $y_1 = x^2$ solves $x^2y'' + 2xy' - 6y = 0$, use reduction of order to find a second solution $y_2(x)$.

P4. *Characteristic Equation — Repeated Root (IVP)*

Solve the initial value problem.

$$y''' + 6y'' + 9y' = 0, \quad y(0) = 0, \quad y'(0) = 1, \quad y''(0) = -4.$$

P5. *Characteristic Equation — Higher Order*

Find the general solution of

$$\frac{d^4 y}{dx^4} - 14 \frac{d^2 y}{dx^2} - 32y = 0.$$

P6. *Undetermined Coefficients — Superposition with Modification Rule (IVP)*

Solve the initial value problem.

$$y'' + 4y' + 4y = (7 + x)e^{-2x}, \quad y(0) = 6, \quad y'(0) = 8.$$

P7. *Undetermined Coefficients — Resonance (IVP)*

Solve the initial value problem.

$$y'' + y = 10 \cos(2x) - 4 \sin(x), \quad y\left(\frac{\pi}{2}\right) = -1, \quad y'\left(\frac{\pi}{2}\right) = 0.$$

P8. *Annihilator — Identify the Operator*

Find a linear differential operator that annihilates each function.

(a) $(7 - e^x)^2$

(b) $8 + e^x \cos(5x)$

(c) $e^{-x} \sin(x) - e^{7x} \cos(x)$

P9. *Annihilator — Full Solution*

Use the annihilator method to find the general solution of

$$y'' + 3y' - 4y = 16e^{2x}.$$

P10. *Mixed Review — Third-Order Equation*

Find the general solution of

$$3y''' + 7y'' + 10y' + 8y = 0.$$

Hint: test $r = -4/3$.